* Anomaly Detection : [To detect Outliers].

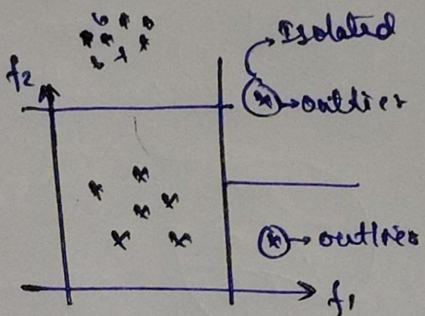
↳ Play an important role

[IPL]

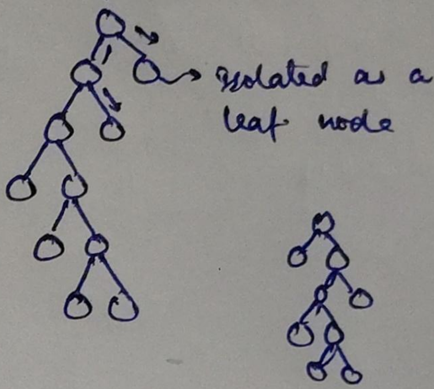
1	15
2	10
3	12
$\boxed{4}$	$\boxed{100}$ > 36

1. Isolation Forest [Decision Tree]

f_1	f_2	f_3	f_4
–	–	–	–
–	–	–	–
–	–	–	–
–	–	–	–



Isolated Tree



Anomaly Score

Mathematical formula : Compute anomaly score for a new point

$$\boxed{S(x, m) = 2^{\frac{-E(h(x))}{c(m)}}}$$

m = no. of data points

x = Data point

$E(h(x))$ = Average search depth for x from the isolated tree.

$c(m)$ = Average value of $h(x)$ [Threshold ≥ 0.5]

$E(h(x)) << c(m) \Rightarrow S(x, m) \approx 1 \Rightarrow$ Average score $\Rightarrow$ outlier

$E(h(x)) >> c(m) \Rightarrow S(x, m) \approx 0.5 \Rightarrow$ Normal data point.